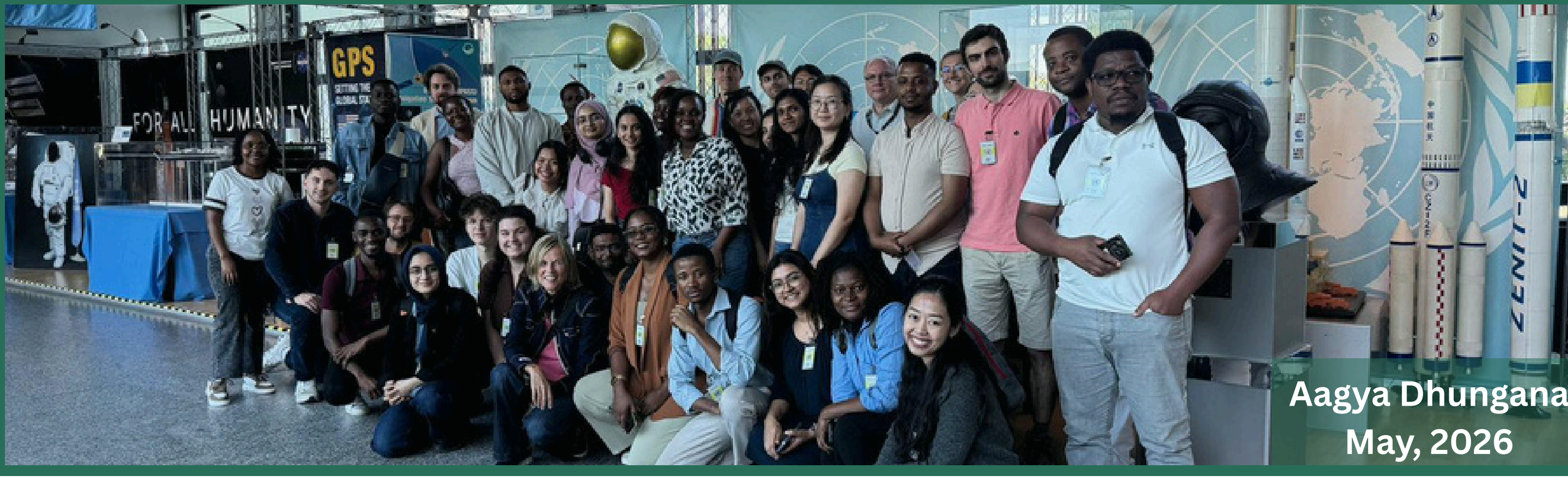

Copernicus Hubs and Institutions – Excursion to UNOOSA and EODC in Vienna

On 26th May 2026, our CDE class packed up for an educational excursion to Vienna, and honestly, it turned out to be one of those days that makes you fall back in love with your geospatial field. I was excited to see geospatial knowledge gathered throughout the semester move from theory to practice. We walked into the buildings where data gets stored, managed, and turned into decisions, and policies. We visited two foundational players in the geospatial industry and got a front-row seat to how international organizations actually function behind the scenes. What follows is everything we learned from both stops; Earth Observation Data Center(EODC) and United Nations Office for Outer Space Affairs(UNOOSA), the questions that came up along the way, and a few thoughts that I put together on my way back till the time I wrote this down.

Earth Observation Data Centre (EODC)

Our first stop of the day was EODC. We left Salzburg at the very early hour of 6 AM (Probably the first time I woke up that early after coming to Austria) and rolled into Vienna around 10 AM, just in time to be welcomed by Christian Briese, Managing Director of EODC, who kicked things off with an orientation before walking us through the infrastructure exhibition. This is where the theory finally met the metal (quite literally to cool down the HPC) we got to see, with our own eyes, how Earth Observation data is actually stored and how the whole system of spatial data infrastructure(SDI) functions. Mr. Briese explained that EODC has been guided by the principle of open and interoperable data since it was founded in 2014, and that this principle is really the engine behind everything else they do. By building distributed IT infrastructure, EODC has been strengthening collaboration between the public and private sectors, making sure data is accessible to everyone- not just the institutions lucky enough to have their own supercomputers in the basement.

Of course, "open and accessible for everyone" sounds simple until you realize what it actually takes to make that happen: high-capacity storage, serious hardware and ALOT of cooling energy, and need for physical space. EODC's premises also houses a high-level supercomputer managed by the Austrian Scientific Cluster (ASC), which supports EODC's mission directly. In short, EODC runs on a combination of multiple data infrastructures and organizations working together as a system— cloud storage, high-performance computers, and a steady stream of data ingestion from Copernicus sources. Standing there, we got the full picture of where our data was stored in real life.

HPC Storage Unit (The full picture)

We also learned that ASC, previously known as the Vienna Scientific Cluster, installed one of the most powerful supercomputers in Austria and is currently in the process of upgrading it even further. On top of that, ASC is also behind MUSICA, a project supporting distributed processing and computation. I had only learnt about this concept in theory but now I see it functional where the HPC and storage systems are spread across three locations — Vienna, Innsbruck, and Linz . This distributed setup that lets multiple organizations collaborate and, just as importantly, gives the whole system a backup if one location goes down. Distributed computing sounded great on paper, but when we asked about the challenges behind it, we learnt it came with a real cost: more infrastructure, more cooling, and a bigger bill at the end of the month.

Alongside the projects already running, we also got a sneak peek into what's next for EODC, and it was genuinely exciting to know about "AI Factory Austria," which will host AI-related projects and AI-optimized supercomputers. More exciting was to learn that it was a part of a bigger network of AI factories being built across Europe to support collaboration in the AI space. It's a nice reminder that Europe is trying to build the infrastructure that will be most relevant in present as well as in future.

Key takeaways

- I learned about the backend of EO data for the first time. I'd only ever known the front end — download data, analyse it, get a result — without ever thinking about where that data physically lives or what organizations keeps it running.
- Distributed computing is often praised as the ideal setup for backup and resilience, but in practice, the cost and energy demands of actually implementing it are high. The benefits are real, but so is the price tag and effort.
- Europe is moving fast to keep up with AI, and the AI Factory initiative is proof of that — though there's clearly still room to grow in multi-disciplinary cross collaboration.

United Nations Office for Outer Space Affairs(UNOOSA)

UN Vienna grounds

Interesting enough, this visit had been brewing in my head for 10 months before it actually happened. During my first big Earth Observation class, I'd asked Martin (my Professor) a question about the guiding principles for launching EO satellites into space, and that's when he first mentioned UNOOSA — the body responsible for such laws. Ever since that day, I'd been quietly excited to one day visit the organization myself to ask the question. So when I finally got the chance to ask a question during our session at UNOOSA, I didn't even think twice , I asked the exact same question that had been sitting on for 10 whole months: why isn't there a clearer law, and how is all of this actually being regulated? And I got the most honest, refreshingly direct answer from Lorant

Czaran, Chief of the Space Applications Section at UNOOSA. Getting to learn straight from him, without any filter, what a privilege it was.

For context, UNOOSA is the UN's regulatory division looking after projects that support, promote, and implement space applications. Its work spans an enormous range — from helping small villages in Africa develop and launch their own small satellites, to drafting the regulations and policies that keep remote sensing projects safe, open, and accessible to everyone. We got to learn about everything UNOOSA has been doing since it was first established in 1958 (and yes, the internet gives you a handful of different dates depending on where you look, but I'm sticking with the one on Wikipedia). What stood out most was how UNOOSA has kept pushing forward on social development despite working with limited funding, which honestly made the whole thing more inspiring. One example that really stuck with me: UNOOSA has been supporting undergraduate students who want to experiment with building small satellites but don't have the funds to do it — and they've kept this support going for years, just so these students' interest in the space ecosystem doesn't die out because of money problems. And I believe this form of advocacy is built on actual trust.

Beside such projects, among many, UNOOSA also runs its own disaster information geo-dashboard, which provides near real-time disaster risk assessments — and I noticed quite a few incidents reported from Nepal, my own country. Seeing that was a small but powerful reminder that UNOOSA's work genuinely reaches the most disaster prone and vulnerable places that need it most.

Now, circling back to my question — and the answer I will probably remember for a long time. Czaran broke it down in the simplest way possible: the guiding principles don't legally bind organizations, they just tell them how things are supposed to be done. So the companies launching satellites are the ones responsible for following these guidelines, and if they choose not to, UNOOSA legally can't do much about it. In simpler words: rules exist, but enforcing them depends entirely on whether countries and companies actually choose to follow them. And look, I won't pretend I didn't think "is this just capitalism doing its thing again?" — but I'll leave that thought right there before this report turns into a different kind of essay. Either way, I got my answer, and I walked out of that room buzzing with motivation to build my own dashboard tracking the number of satellites in space alongside their relevant policies, so anyone can look at it and understand what's actually going on up there at a glance.

Key takeaways

- UNOOSA's continued support for small satellite development among students and small communities, with limited big budgets, is one of the most genuine forms of space advocacy And I walked out more inspired about my field than ever to know they keeps curiosity alive instead of letting funding gaps shut it down.
- Satellite congestion and "space pollution" is a real and current concern and the gap between having guiding principles and actually enforcing and adapting them is exactly why space debris are growing exponentially.

Me with stone from the Moon

Parting note

Excursions like this enables us to sit across from professionals and experts to discuss current issues, soak up motivation for our own ideas, and leave with our heads full of new things to build. Personally, this trip was a genuine source of inspiration, and I really hope CDE keeps creating and even expanding opportunities like this to collaborate and talk directly with field experts. I'm deeply grateful to CDE for this opportunity to learn things I'd never had access to before this trip.